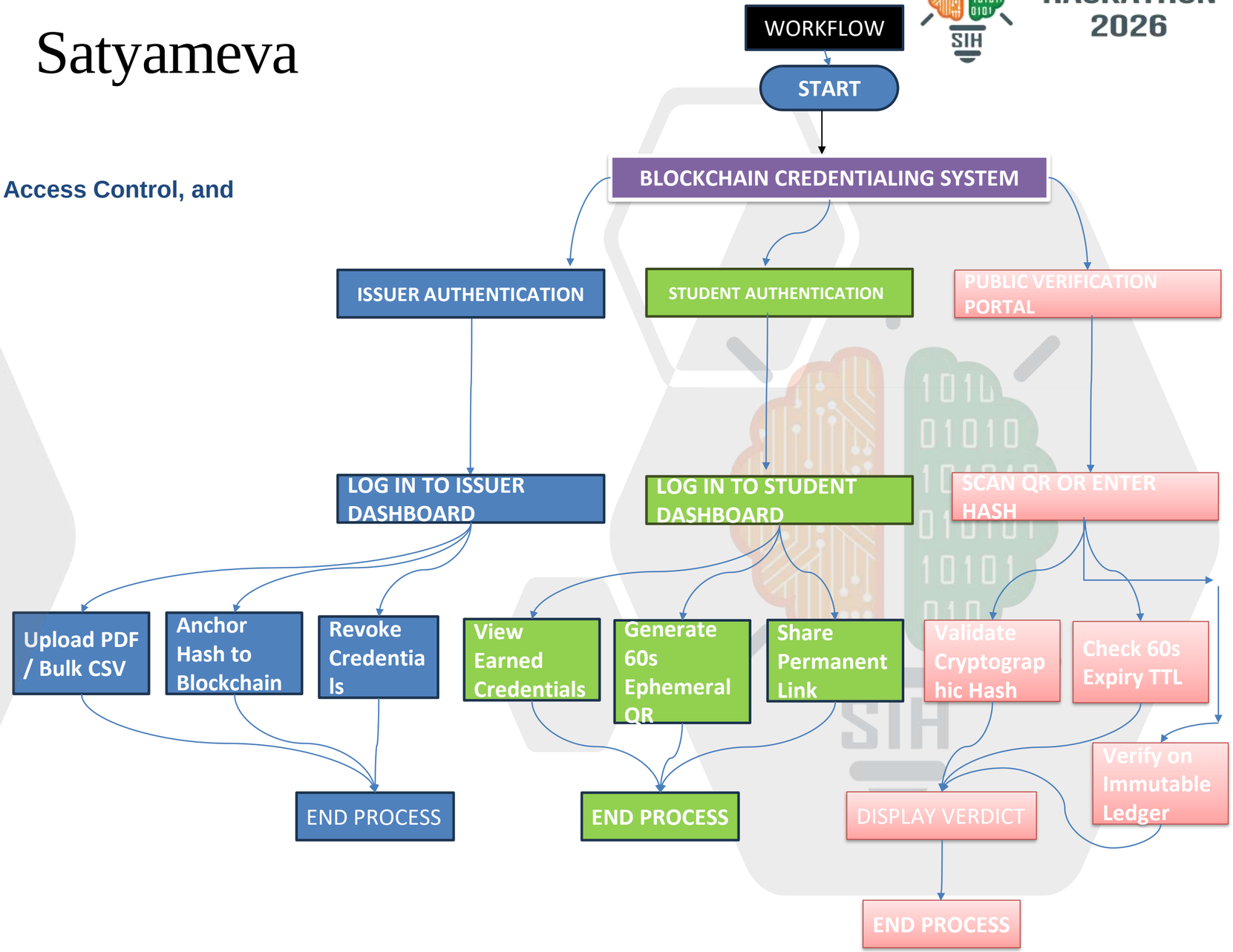
SMART INDIA HACKATHON 2026

Satyameva

- **Problem Statement ID:-** SIH26125
- **Title:-** Blockchain-Based Secure Platform for Identity, Access Control, and Digital Asset Management
- **Theme:-** Blockchain & Cybersecurity
- **Category:-** Software
- **Team ID:-** 10
- **Team Name:-** CYPHER-X
- **Team Members :-**

A Praneeth Sai (Team Leader, CSE-CS)	Jai Chauhan (Team Member, CSE-CS)
Jyoti Rana (Team Member, CSE)	Gauri (Team Member, Biotechnology)
Sparsh (Team Member, CSE)	Akshit Thakur (Team Member, CSE-AI&DS)



CYPHER-X

Satyameva

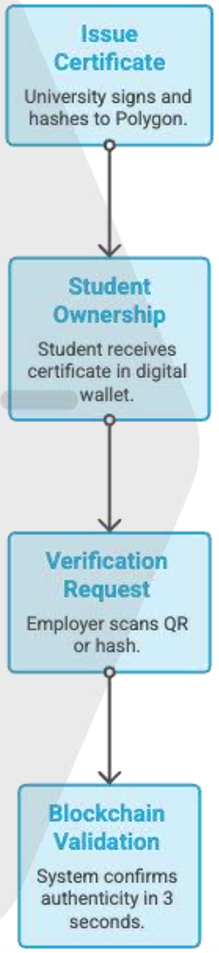
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Blockchain based certificate issuance and instant verification replacing weeks of manual checks and fake degree fraud with a 3 second, tamper-proof QR scan.

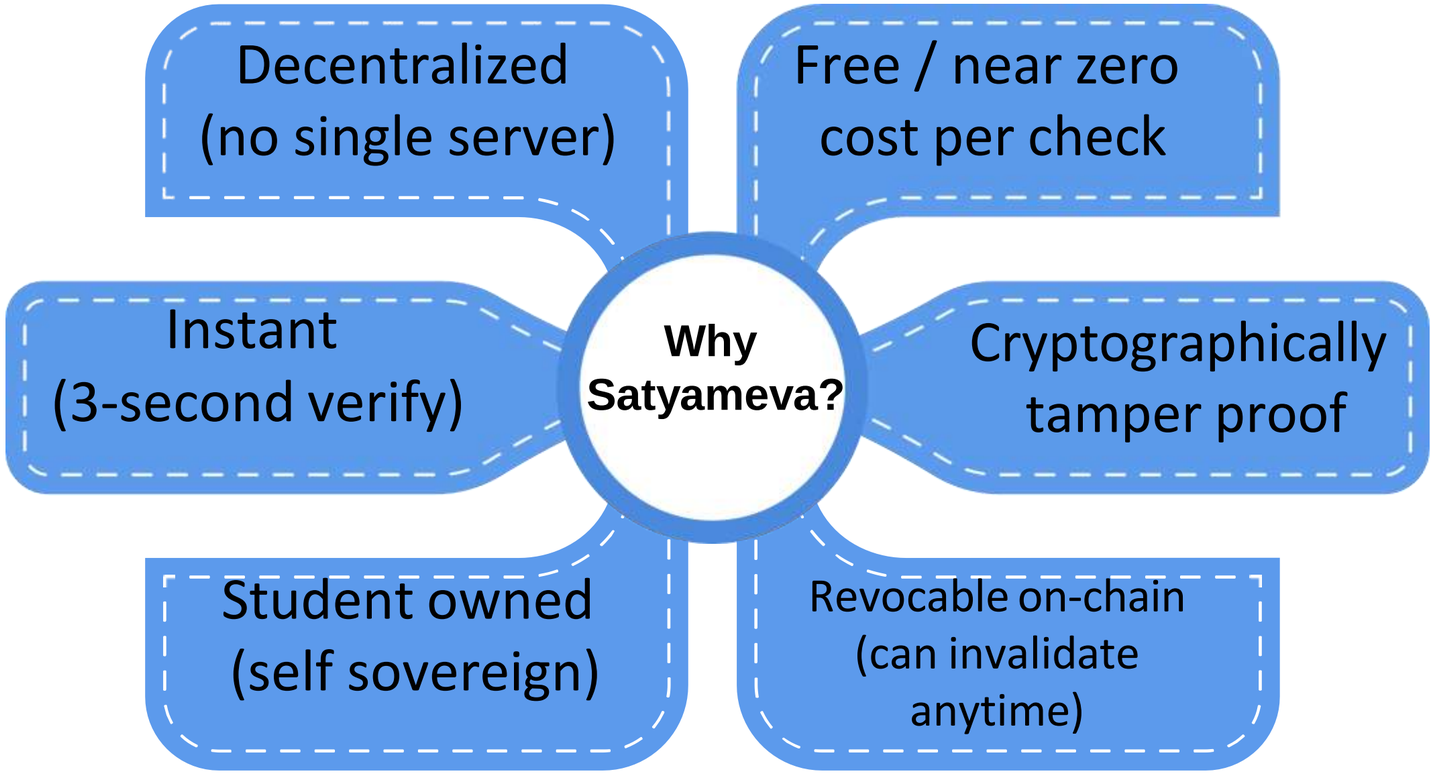
What is Satyameva?

Satyameva transforms certificates into tamper proof blockchain credentials that students own and anyone can verify instantly.

Blockchain Certificate Lifecycle



Whats the USP?



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TECHNICAL APPROACH

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Techstack and Ui mockups

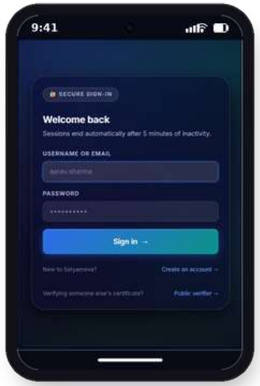
Frontend
React.js/ HTML+js

Backend
Node.js + Web3.js

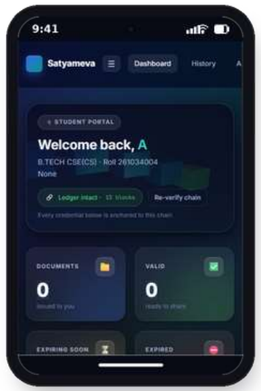
Database
SQLite, SQLAlche-my ORM

Security
SHA-256, HMAC-SHA256, PBKDF2

Blockchain
Custom PoW engine



Secure Sign-In



Student Dashboard

Methodology

Synchronous Processing → Cryptographic Tamper Proofing → Seamless Notifications

Process/ Implementation flow

1. University fills certificate details & signs digitally (private key)
2. Smart contract generates SHA-256 hash & mints it on Polygon
3. Certificate PDF is AES-256 encrypted & stored on IPFS
4. Student receives certificate + QR code in digital wallet
5. Employer scans QR / instant on-chain verification (~3 sec)

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FEASIBILITY AND VIABILITY

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~2sec
blockchain transaction / certificate mint time

₹0.08
Cost per certificate on Polygon network

₹8Lakh
total cost to issue 1 crore (10M) certificates

Phase Rollout Roadmap

Phase 1
Pilot at own college

→

Phase 2
state university network

→

Phase 3
UGC mandate+ SMS lookup

→

Phase 4
National scale
(40M+ students)

Risk and mitigation strategy

- Key compromise → Multi-sig wallet (2-of-3) + on-chain revocation
- No smartphone access - Printable QR + Phase 3 SMS lookup
- Adoption resistance → Pilot in-house, then push for UGC mandate
- DigiLocker overlap → Complementary trust layer, not a replacement

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IMPACT AND BENEFITS

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40,000+
fake degrees officially caught in India every year (reported cases only)

~99.99%
faster verification: 2–4 weeks → 3 seconds

~99.99%
cheaper per check: ₹500–₹2,000 → ₹0.08

40M+
higher-education students in India (addressable scale)

Key Stakeholder Impact

- **Students:** Lifetime access to a secure, tamper-proof digital wallet. Eliminates the anxiety and risk of losing physical documents.
- Protects institutional brand reputation from counterfeit degrees. Enjoys zero gas fees for mass bulk issuance.
- Achieves 100% cryptographic certainty instantly. Eliminates the friction of waiting weeks for manual background checks.

Benefits Created

1. Social:Restores trust in degrees; protects genuine graduates from fraud stigma.
2. Economic: Cuts verification fees saves ~₹500-2000 Cr at 1 crore certificates.
3. Environmental: Fully paperless no mailers or couriers between institutions..
4. Policy: Aligned with NEP 2020 and Digital India.

RESEARCH AND REFERENCES

1	UGC Fake Universities Data & Reports	Purpose: Validates the core problem statement by providing official government data on the rampant issue of counterfeit degrees and unapproved institutions in India.
2	NIST FIPS 180-4 (SHA-256 Cryptographic Standard)	Purpose: Validates the core technical decision to use SHA-256 for payload integrity, mathematically ensuring that credential tampering is impossible to hide.
3	RFC 2104: HMAC (Hash-based Message Authentication)	Purpose: Formalizes the standard for HMAC, validating our approach of using secret keys and ephemeral timestamps (60s TTL) to authenticate QR codes and prevent replay attacks.
4	Blockcerts: Open Standard for Blockchain Credentials	Purpose: Demonstrates the industry-accepted standard (from MIT Media Lab) for anchoring academic records to an immutable ledger for decentralized, tamper-proof verification.
5	Flask Framework & SQLite Ecosystem Documentation	Purpose: Validates our technology stack choice, proving that a lightweight, serverless SQLite blockchain paired with Flask is highly feasible and cost-effective for institutional deployments.